Project Report

Road Proposal

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Introduction

In this project, we adopted to apply a generalized approach to the Portuguese regulations concerned with road design. Two variants are proposed for roads that are aimed for local transit, but due to its location regarding the town (peripheral), we decided to adopt national roads criteria, also for learning and assimilating purposes.

The town where this road should be located is Boticas, a town with ~1000 inhabitants, in the portuguese northeastern countryside, known for its landscapes.

Basic Elements

Road Category

At present, the Portuguese Road Network comprises 2 road categories, that constitute the **Fundamental National Network** and the **Complementary National Network**.

The Fundamental National Network is constituted by the Main Itineraries (*Itinerários Principais*, IP). The Complementary National Network integrates the Complementary Itineraries (*Itinerários Complementares, IC*) and the National Roads (*Estradas Nacionais, EN*).

Though the road designed is meant for local traffic, we considered it to be a National Road, due to its distance to the center of the town, and the speeds that may be adopted by the drivers.

Landscape Integration

It is important to any road that its appearance is pleasant and enjoyable. Consequently, the aspects of landscape integration must be considered together with safety and all the other factors in the designing of a road. This is particularly important when roads are located in areas of acknowledged natural beauty. The horizontal and vertical alignments, along with the cross-section must harmonize with the environment, justifying a reasonable raise in costs, to highlight the beauty of its path and surroundings.

The following aspects of the Portuguese Designing Norm[1] were considered, in the course of the execution of this project:

1. The tracing implantation must enable the preservation of the environment and the view of landscapes;
2. The horizontal alignment must integrate in the character of the traversed area, in a way that the “wounds” caused by the excavations and landfills are the least possible;
3. The destruction of existing vegetation must be minimized. In practice, an inventory of the affected tree must be performed, in order to proceed to their replantation;
4. The sloped must be softened when possible and coated with bushes, in way that they integrate perfectly with the surroundings.

Speed

Design Speed

The **design speed** is arbitrated in the elaboration of a road project, which conditions some of the geometric characteristics of the road. Choosing the design speed is mainly connected with the road category and function in the national network, considering topographical, environmental and economical conditions. The design speed allows the selection of the majority of the geometric characteristics to consider in the project, and thus it contributes to standardize its design. In Table I, the different design speed to be considered are represented:

Road Type		Design Speed (km/h)				
		140	120	100	80	**60**
Two carriageways	IP	X (a)	X (a)	X	-	-
	IC	X (a)	X (a)	X	X (b)	-

One carriageway, with two directions	IC	-	-	(c)	X	
	EN	**-**	**-**	**-**	X	**X**

(a) Only in highways
(b) This case must be justified
(c) Only in the cases where future duplication is expected

Table 1 - Design speed values

The design speed is used to determine the minimum horizontal radius, the maximum gradient of the vertical alignments and the type of the cross-section.

It must be constant during the course of the road, contributing for the desirable standardization of the geometric characteristics.

For this project, we considered the generalized design speed of **60 km/h**.

Traffic Speed

From a cumulative speed distribution curve in a road section, it is possible to determine, for each speed, the percentage of vehicles (percentile) circulating at or below the same speed. The speed corresponding to the 85th percentile, which is taken as the traffic speed, is usually considered to be at a critical speed, since small increases in the percentile from the 85th percentile correspond to large increases in speed and speeds higher than these are usually dangerous to the conditions considered.

In Table 2, the traffic speed values corresponding to different design speed values are represented:

Design Speed (DS)	Traffic Speed (TS)
40	50
50	60
60	**80**
70	90
80	100
90	110
100	120

110	125
120	130
130	135
140	140

Table 2 - Traffic speed values in the Portuguese Road Network (km/h)

It can therefore be said that in defining the geometric characteristics of a road, it is necessary to consider not only the design speed but also the traffic speed, which corresponds to the use of a more realistic concept from the dynamic point of view, since the speed practiced by the users along the route varies according to its characteristics. Hence, it seeks to satisfy the expectations of the drivers. Table 3 refers to the standard speed to be considered in the definition of different geometric characteristics of the alignment.

Alignment Elements	Speed	
	Design Speed	Traffic Speed
Minimum horizontal radius	X	-
Maximum gradient	X	-
Type of cross-section	X	-
Visibility distances	-	X
Minimum radius of the vertical curves	-	X

Table 3 - Speed values to consider in the determination of various alignment elements

The generalized traffic speed adopted in this project was **80 km/h**.

Visibility

Stopping sight distance

The **stopping sight distance** (SSD) is the minimum distance required for a driver, circulating at a certain speed, to safely stop the vehicle, if necessary, after seeing an obstacle on the road

surface. The stopping sight distance is the minimum sight distance that must be ensured when it is not economically feasible to ensure a higher sight distance.

The stopping sight distance is measured between the eyes of the driver (1.05 m above the pavement) and an obstacle on the pavement with 0.15 m in height.

It is determined by adding two plots, the first translating the distance traveled during the driver's perception-reaction time (taken as 2 s), and the second the braking distance that is a function (TS in km/h), the coefficient of longitudinal friction (f_l) and the gradient (+ i or - i, up and down respectively, expressed in % / 100). The following expression is used for the SSD calculation:

$$SSD = \frac{TS}{1{,}8} + \frac{TS^2}{254(fl \pm i)}$$

In this case, the minimum SSD is **120 metres**, with a f_l of 0,32.

Through Autodesk Infraworks, a stopping sight distance analysis was made for the traffic speed of 80 km/h, with the following results:

Figure 1 - Variant 1 Stopping Sight Distance Analysis (Autodesk Infraworks)

Figure 2 - Variant 2 Stopping Sight Distance Analysis (Autodesk Infraworks)

Passing Sight Distance

The stopping sight distance is not sufficient to ensure efficient operation of road users in all circumstances. In fact, the stopping sight distance is inadequate, for example when drivers have to make a complex decision, or when the information given by the signaling is not immediately understood.

The sight distance required for a driver to perceive unexpected, less easy to understand information or a change in the characteristics of the road, to identify that situation, to decide and adopt the most convenient speed and to start and conclude safely the necessary maneuver, is referred to as **passing sight distance**.

It is therefore necessary to ensure the passing sight distance in places where the driver's expectations can be significantly altered or where it is likely to be doubts of the driver when receiving information.

In this case, the level crossings, the connecting nodes, the reduction of the number of lanes, the crossing areas, the service areas and the similar facilities to support the users.

The following expression is used for the PSD calculation:

$$PSD = 3,3 \cdot TS$$

In this case, the minimum PSD is **270 metres**.

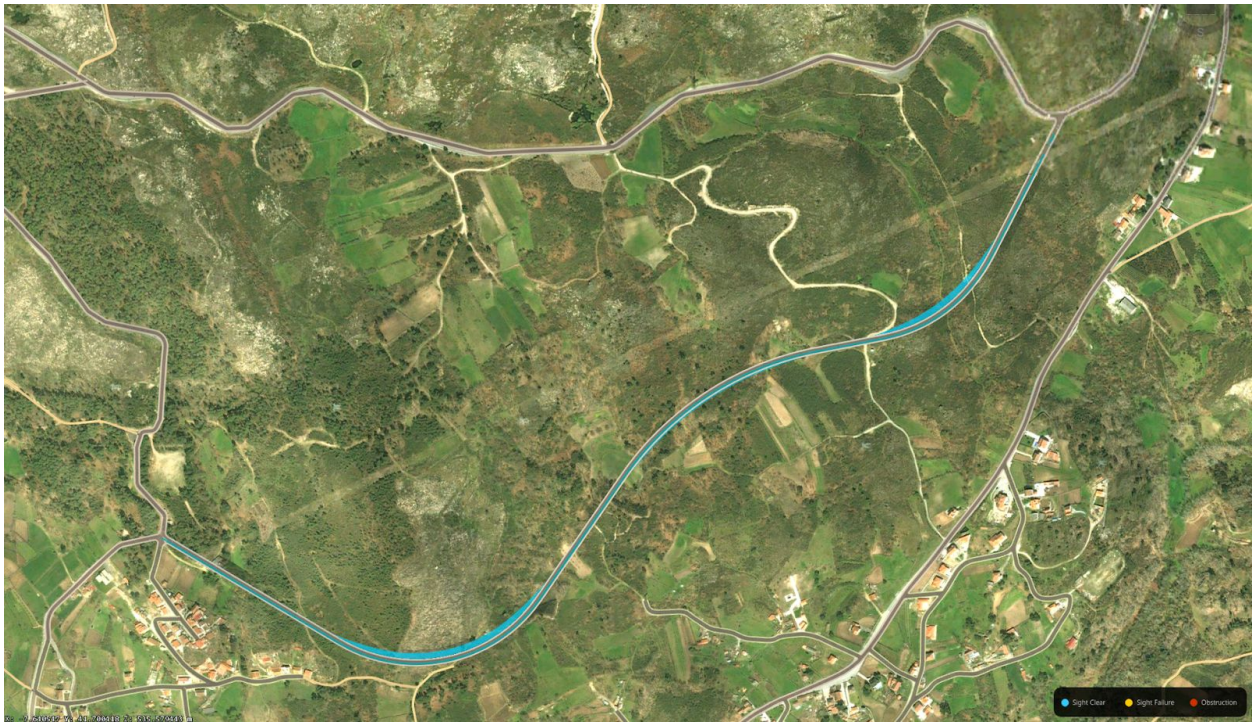


Figure 1 - Variant 1 Passing Sight Distance Analysis (Autodesk Infraworks)

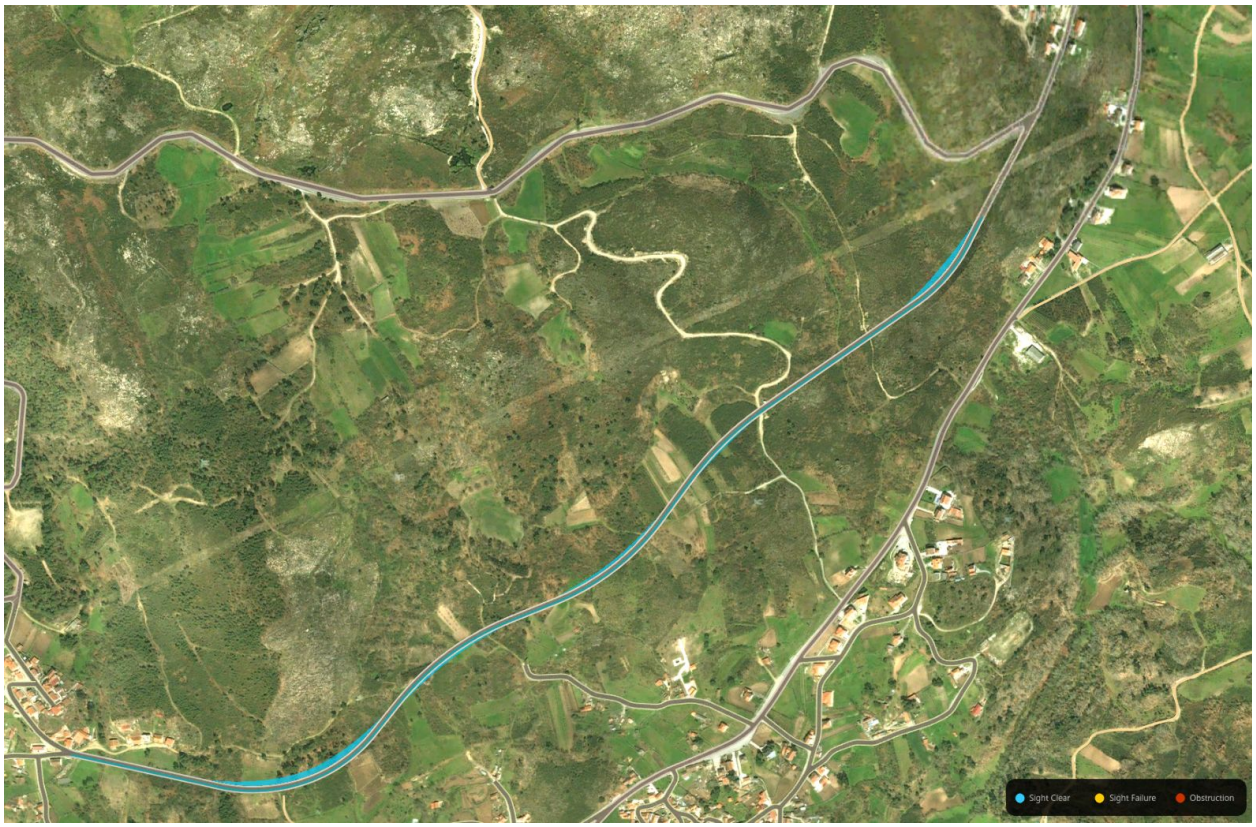


Figure 2 - Variant 2 Passing Sight Distance Analysis (Autodesk Infraworks)

Geometry

Horizontal alignment profile

Straight alignments

Extensive straight alignments make it easy to overtake in two carriageways roads. However, they usually integrate poorly in topography. In addition, they increase the duration of dazzling during night driving, make driving more monotonous, and make it difficult to assess speeds and distances.

Extensive straight alignments should not have constant longitudinal inclinations. In order to reduce night-time dazzling and driving monotony, the maximum extent of a straight-line alignment (in metres) with constant longitudinal inclination shall be 20 VB, with VB being the base speed in km/h.

Straight alignments should also provide for the possibility of overtaking so that on single carriageways, with two directions, the recommended minimum length (LR_{min}, in metres) is 6DS (DS in km / h) - Table 4.

Extension (m)	Design Speed (km/h)										
	40	50	**60**	70	80	90	100	110	120	130	140
Minimum (6DS)*	240	300	**360**	420	480	540	600	660	720	780	840
Maximum (20DS)**	800	1000	**1200**	1400	1600	1800	2000	2200	2400	2600	2800

* Not applicable to bi-directional roadways ** Applicable on steady inclines

Table 4 - Minimum and maximum extensions of straight alignments - Indicative values

Whenever possible, orientations of straight alignments coinciding with the east-west direction should be avoided in order to avoid dazzling caused by the sun, especially in the case of extensive alignments.

None of the variants has an extensive straight alignment with the following criteria due to the difficult topography at hand.

Circular curves

Minimum absolute radii (AR)

When a vehicle traverses a curve it is subjected to a centrifugal force (F), whose intensity is given by:

$$F = \frac{M \cdot V^2}{R}$$

in which M represents the mass of the vehicle, V the speed and R the radius of the curve.

The resistance to this centrifugal force is ensured by the transverse frictional force mobilized at the tire-pavement interface and by the weight component of the vehicle, in the opposite direction to the centrifugal force, generated by the introduction of the overhead lift. In this situation, the equilibrium condition in relation to the lateral slip, allows to deduce the following relation:

$$R = \frac{V^2}{127(ft + Se)}$$

in which:

V speed (km/h)

R	curvature radius (m)
S_e	superelevation (%/100)
f_t	tangential coefficient of adhesion

The coefficient of friction between the tires and an ice deck on their surface is about 0.1, which means that in this case a vehicle standing or moving slowly tends to slide to the center of the curve when overtaking is of the order of 10%. Consequently, in several European countries, overhead is currently limited to a maximum of 7%. By assigning a maximum value to the coefficient of transversal friction, corresponding to each speed (Table VII), and the value of 7% to the overhead, the absolute minimum radii are determined (Table 5).

Design Speed (km/h)	f_t
40	0.16
50	0.16
60	**0.15**
70	0.14
80	0.14
90	0.13
100	0.12
110	0.10
120	0.09
130	0.08
140	0.06

Table 5 - Coefficient value of transversal grip

In this particular case, the coefficient adopted was **0.15** and the maximum superelevation was **7%**. These radii must be used exceptionally since their use results in high-value centrifugal accelerations which, for base speeds of less than 80 km/h, are close to the maximum permissible for safety and comfort.

Minimum normal radii (NR)

Normal minimum radii shall ensure safe and comfortable circulation and they shall be such that the centrifugal acceleration, in the case of a vehicle at design speed, has a value corresponding to 50% of the maximum permissible value, taken as 0.22g , for the purpose of this standard.
Table 6 shows the values of normal minimum radii, which should be considered in the projects, except in special duly justified cases.

Design speed (km/h)	Minimum Absolute Radius (m)	Minimum Normal Radius (m)
40	55	110
50	85	180
60	**130**	**250**
70	180	350
80	240	450
90	320	550
100	420	700
110	560	850
120	700	1000
130	900	1200
140	1200	1400

Table 6 - Horizontal minimum radii

The minimum radii considered for this projected were **130 metres for the absolute radius** and **250 meters for the normal radius**.
In Variant 1, the following circular curve was projected:

Transition curves

Among the possible transitional curves, the clothoids must be adopted, due in particular to comfort and safety issues, since a vehicle circulating along this type of transition curve, from straight alignment to circular curve, at constant speed, is subject to uniform variation of the centrifugal acceleration.

The extent of the two transition curves associated with a circular curve should, where possible, be between 1/2 and 2/3 of the total development of the curved alignment.

Table 8 indicates the minimum value of the parameter of the clothoid (A) as a function of the design speed. The value in bold were the ones generally adopted in this project.

Parameter	Design speed										
	40	50	**60**	70	80	90	100	110	120	130	140
Minimum normal radius (m)	110	180	**250**	350	450	550	700	850	1000	1200	1400
Minimum extension of the curves (3DS)	120	150	**180**	210	240	270	300	330	360	390	420
Minimum parameter of the clothoid (m)	52	73	**96**	121	150	183	233	283	333	400	466
Minimum radius without superelevation (m)	**≥ 2500**							≥ 5000			

Table 8 - Horizontal curves parameters

Where it is not possible to ensure minimum development of circular curves, they should be made known as soon as possible, such as in straight straight lines, and in this case the choice of a reduced clothoid parameter (A = R/3) is appropriate. The transition is thus less smooth from the optical point of view, inducing sensation of greater danger; warned drivers slow down.

Other aspects

Straight alignments before and between curves

The minimum radii of the circular curves must still be related to the extension of the straight alignments that precede them, in order to ensure a homogeneous trace. The maximum and minimum lengths of the straight alignments are defined according to the certain criteria.

If the length of the straight alignment (LR) is less than the value of its calculated minimum length (LRmin), what is relevant for safety is the relationship between the unconstrained velocities ($V_{1,85}$ and $V_{2,85}$) of the consecutive curves and not that length, according to Table 7. $V_{1,85}$ is considered to be the unimpeded speed on the curve immediately upstream of the straight alignment and $V_{2,85}$ is the unimpeded speed on the curve immediately downstream of the straight alignment.

$V_{1,85}$ (km/h)	$V_{2,85}$ (km/h)								
	50	60	70	80	90	100	110	120	130
60	**50**								
70	109	**59**							
80		127	68						
90			145	77					
100				163	86				
110					182	95			
120						200	104		
130							218	113	
140								236	123

Table 7 - Minimum length of a straight alignment (LR_{min}) as a function of the unimpeded speed

Vertical alignment profile

Gradients

Maximum slope

In the gradients with a large slope and considerable extent, attention should be paid to the reduction of speed, especially of heavy vehicles, resulting in a decrease in the level of service and an increase in operating costs. Consideration should also be given to the need for longer braking distances on descents.

The maximum slope of the gradients shall be as shown in Table 8.

Gradients with the maximum slope referred to in this table must not extend more than 3 km, even if additional lanes are built to overtake slow vehicles.

In the case of rough terrain, the limits referred to in Table 8 may be increased by 1% on gradients with a maximum length of 600 metres and 2% on gradients with a maximum length of 300 metres.

Design speed (km/h)	Maximum desirable slopes (%)
40	8
60	**7**
80	6
100	5
120	4 (a)
140	3

(a) In highways, the maximum inclination must be 3%.

Table 8 - Gradients maximum slopes

The maximum slopes referred to in Table 8 are satisfactory for most situations and usually allow for economic earthmoving. However, in the case of rough terrain, if the implications of the maximum slopes in the cost of construction and in the environment are too severe, a comparative economic study of construction and environmental costs with operating costs. In general, economic studies show that while there are clear burdens for users with higher than desirable slopes, the cost increase resulting from a smoother profile is normally justified only for very high traffic volumes, which is not the case for the project at hand. This way, it was opted to reduce earthmoving as much as possible.

Minimum slope

Gradients with a very small slope must be avoided, especially when they coincide with a very small superelevation. The minimum gradient slope of 0.5% should assure the surface water drainage.

In curved areas, the slope of the gradient should be greater than the obliquity, ie:

$$i - \Delta i \geq 0.5\%$$

where:

i gradient slope (%)

Δi obliquity, longitudinal inclination of the shoulder of the carriageway comparing to the axis (%)

Critical extension of the gradients

The maximum slope does not by itself ensure the necessary control of the gradients. It is also necessary to consider their extension, since it is their combination which affects the traffic capacity, the level of service and the loss of time of the vehicles. The criterion normally used to define the critical extent of the gradients is based on the reduction in the speed of heavy vehicles compared to light vehicles. Consequently, the critical extension of a gradient is the maximum extension that a gradient can have without causing excessive speed reduction of heavy vehicles.

Gradient slope	3%	4%	5%	6%	7%	8%
Critical extension	420	300	230	180	150	120

Table 9 - Critical extensions of gradients

Vertical curves

Convex curves

The radius of the convex curves is conditioned by the need to ensure the stopping sight distance. There are two hypotheses to be considered regarding the stopping visibility distance, as it may be smaller or larger than the extent of agreement.

Design Speed (km/h)	Traffic Speed (km/h)	Minimum radius		Minimum length	
		Absolute [(a)] (m)	Normal (m)	Absolute [(a)] (m)	Normal (m)
40	50	1500	1500	40	60
50	60	1500	2100	50	60
60	**80**	**2000**	**3000**	**60**	**120**
70	90	3000	4200	70	120
80	100	5000	6000	80	120
90	110	7500	8500	90	120
100	120	9000	12500	100	120
110	125	12000	13000	110	120

120	130	14000	16000	120	120
140	140	20000	20000	140	140

[a] Only acceptable in unidirectional carriageways

Table 10 - Minimum radius of convex curves

This development is intended to ensure optical comfort to drivers. The absolute minimum rays are calculated for an object with a height of 0.60 m.

In particular points of secondary roads (DS equal to 40 km/h or 50 km/h), where the adoption of these minimum values is not economically justified, lower radii may be used, but never lower than those calculated by the formula:

$$R_v = 0.25 \cdot DS^2$$

This could be the particular case of our road, since it is not an official national road (EN).

Concave curves

The development of the concave curve, and consequently its radius, is conditioned by the need to ensure night visibility with vehicle headlights and by comfort (vertical acceleration should not exceed 0.25 m/s^2). The predominant criterion is always night visibility.

Design speed (km/h)	Traffic speed (km/h)	Minimum radius (m)	Minimum length (m)
40	50	800	60
50	60	1200	60
60	**80**	**1600**	**120**
70	90	2500	120
80	100	3500	120
90	110	4500	120
100	120	5500	120
110	125	6000	120

120	130	7000	120
130	135	8000	140
140	140	8000	140

Table 11 - Minimum radius of the concave curves

Cross-sectional profile

Carriageway

Lane width

In terms of geometry, it is considered that the carriageway width is the sum of the lanes width. In national roads with two lanes, these must have the minimum width of 3.5 meters (or 3.0 meters in the case indicated in Table 13). In national roads, lanes with 3.0 meters can be adopted if the design speed is lower than 80 km/h.

Camber

In straight alignements, the camber is normally used to guarantee the surface drainage of the road, since this is the flow direction that ensures the minimum drainage length.

In the roads with two lanes, the pavement is normally inclined to both sides from the axis (inverted “V” profile). This inclination must be of 2.5% for bituminous pavements and 2.0% for cement concrete pavements.

Superelevation

The superelevation of the curves contributes decisively to the safety and comfort of the circulation, as it allows part of the centrifugal force to be compensated for by the force of gravity, as it favors the perception of the curves, and consequently the optical orientation.

It has been shown that in many curves the superelevation is less than desirable. This is because the superelevation is normally set as a function of the base speed, while the drivers travel the curves at the speed they deem compatible with them, which is generally higher than the base speed.

It is concluded that the superelevation should be independent of the design speed and defined according to the unimpeded speed compatible with the radius of each curve.

Consequently, the value of the superelevation, as a function of the radius of the curve, should be that given in Table 12. For the radii with intermediate value, the highest superelevation is adopted.

The line of highest slope on the pavement, which is the result of the combination of the slope of the gradient and that of the superelevation, must not exceed 10%. If this value is higher, the superelevation value indicated in Table 12 should not be reduced. In this case, the slope of the gradient should be reduced or the layout be modified.

For the case of roads with a single carriageway with two ways, we have:

Radius (m)	Superelevation (%)
<525	7.0
525	6.5
600	6.0
700	5.5
850	5.0
1000	4.5
1200	4.0
1400	3.5
1600	3.0
1900 ≤ R ≤ 250	2.5
≥ 2500	-

Table 12 - Superelevation in curves

Superelevation transition

The superelevation transition is done through the transition curve.

For this particular project, the general option offered by Autodesk Infraworks for superelevation was used, as well as its transition, though these should be considered individually, in real practice. The superelevation transition must be studied carefully, since it is particularly important for:

- Assuring a good lateral and longitudinal water drainage;
- Allowing a comfortable variation of the transversal non compensated acceleration by superelevation;
- Enable an adequate optical framing.

Shoulders

The shoulder is an emergency stopping lane for damaged vehicles. They allow the circulation of rescue vehicles, and ensure the lateral support of the carriageway pavement. The shoulders can still prevent an impending accident or reduce its severity. In addition, they increase the traffic capacity of the road. However, for the sake of traffic safety, it is essential that there is a clear distinction between the carriageway and the shoulders, so as to prevent them from being used by the vehicles. Consequently, the shoulders should preferably have a color and texture contrasting with those of the carriageway, as considered in this project.

In addition to the paved shoulder, there should be implemented an unpaved area outside (0.75 m) and the connection between the shoulder and the slope or the gutter (0.60 m).

The width of the paved shoulders should be as given in Table 13. The camber in a straight alignment is that of the lane. In curves with superelevation, the camber of the shoulder is the same as that of the carriageway.

The shoulder of a road must have a constant width, even if the carriageway includes additional lanes, for slow vehicles or for overtaking.

For single carriageways, we have:

Road Type		Design Speed (km/h)	Lane width (m)	Paved shoulders	
				Left (m)	Right (m)
Single carriageway	EN	≥80	3.5	2.5	
	ER	<80	3.0	1.5	

Table 13 - Shoulder width

For this project, the lane width of **3.5 meters** and the shoulder width of **1.5 meters** were considered.

Gutters

Gutters are intended to collect and drive the surface water off the road and should therefore be conveniently sized for the flowrate to be drained.

In general, gutters must be triangular with their bottom at least 0.30 m below the floor level.

The inner side should have a maximum slope (h / b) of 1/4. The outer side may have the slope of the excavation slope.

The slopes (h/b) of the inner and outer sides are always equal to or less than 1/4 and 1/2, respectively.

Comparative Assessment

Horizontal alignment profile

Variant 1

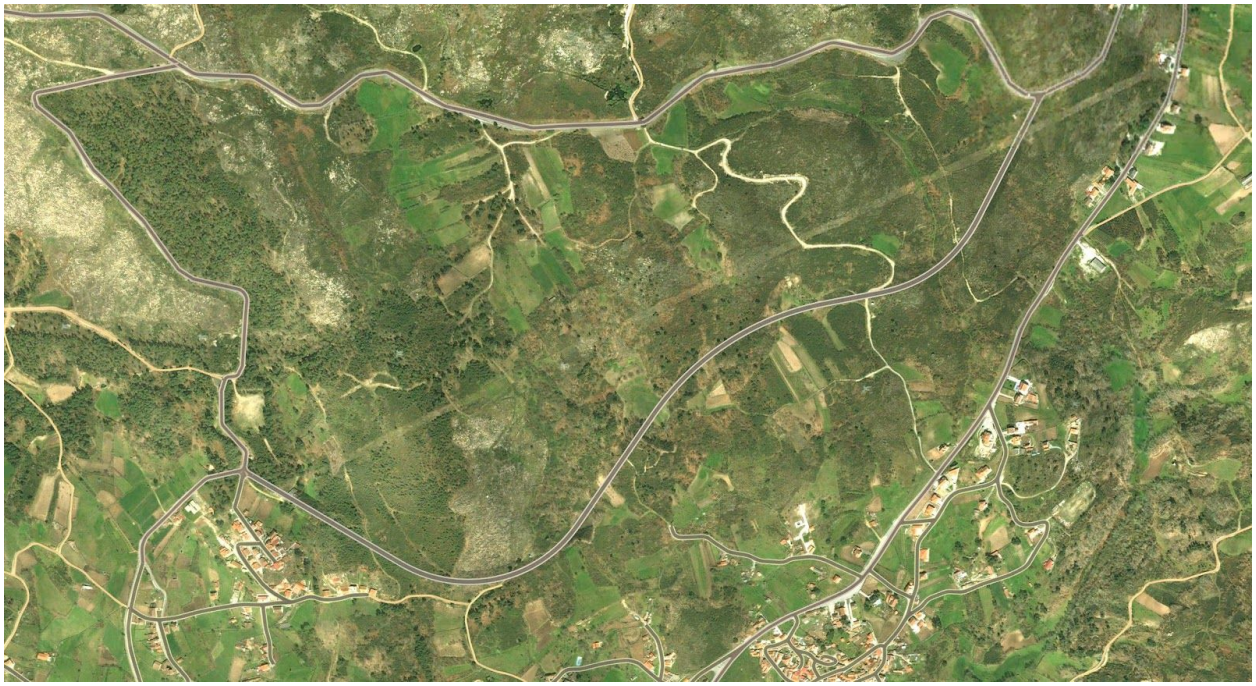


Figure 5 - Horizontal alignment of Variant 1

Variant 1 is composed by the following sequence:

#	Element	Parameters

1	Straight alignment	L = 326.194 m
2	Transition curve	A = 96.177 L = 37.0 m
3	Circular curve	R = 250.0 m L = 342.278 m
4	Transition curve	A = 96.177 L = 37.0 m
5	Straight alignment	L = 235.271 m
6	Circular curve	R = 508.212 m L = 396.140 m
7	Straight alignment	L = 95.753 m
8	Transition curve	A = 37.0 m L = 96.177
9	Circular curve	R = 250.0 m L = 194.459 m
10	Transition curve	A = 37.0 m L = 96.177
11	Straight alignment	L = 238.528 m

Variant 2

Figure 6 - Horizontal alignment of Variant 2

Variant 2 is composed by the following sequence:

#	Element	Parameters
1	Straight alignment	L = 245.064 m
2	Circular curve	R = 250.0 m L = 282.277 m
3	Straight alignment	L = 53.25 m
4	Transition curve	A = 96.0 L = 20.373 m
5	Circular curve	R = 452.372 m L = 209.774 m
6	Transition curve	A = 96.0 L = 20.373 m
7	Straight alignment	L = 68.507 m
9	Circular curve	R = 393.578 m L = 215.131 m
10	Straight alignment	L = 63.806 m
11	Transition curve	A = 133.788 m L = 36.864 m
12	Circular curve	R = 485.548 m L = 124.547 m
13	Straight alignment	L = 277.896 m
14	Circular curve	R = 250.0 m L = 125.227 m
15	Straight alignment	L = 277.939 m

None of the variants has an extensive straight alignment, as the terrain proved to be rather difficult.

In terms of safety and especially when it comes to the dynamism of the road, Variant 2 proves to be a more suitable option, due to the larger number of elements.

Vertical alignment profile

Both profiles were planned manually, without the aid of the *Profile Optimization* tool provided by Autodesk Infraworks, taking into consideration the Portuguese normative.

Variant 1

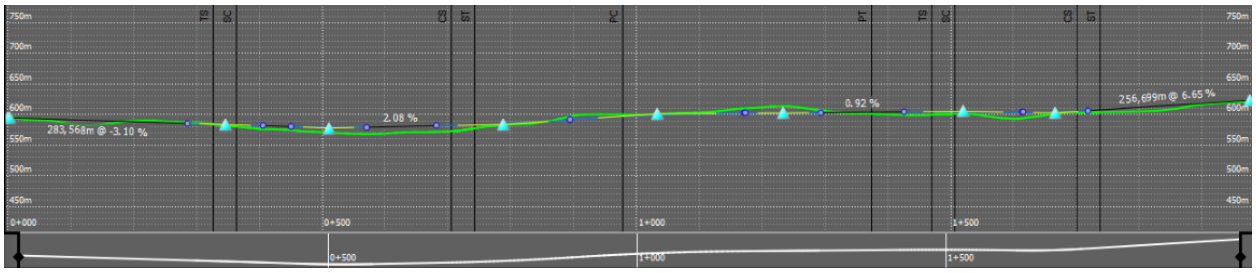


Figure 7 - Vertical alignment of Variant 1

Due to the difficult topography of Variant 1, it is deemed that to execute this profile will oblige a lot of earthmoving efforts. We did not take into consideration the construction of tunnels and bridges, since the road and its traffic doesn't justify it.

The maximum gradient considered was 6.65% and the minimum 0.92%.

Cut	68548.528 m^3
Fill	188649.545 m^3
Net Fill	120101.018 m^3

Table 16 - Variant 1 earthwork quantities

Variant 2

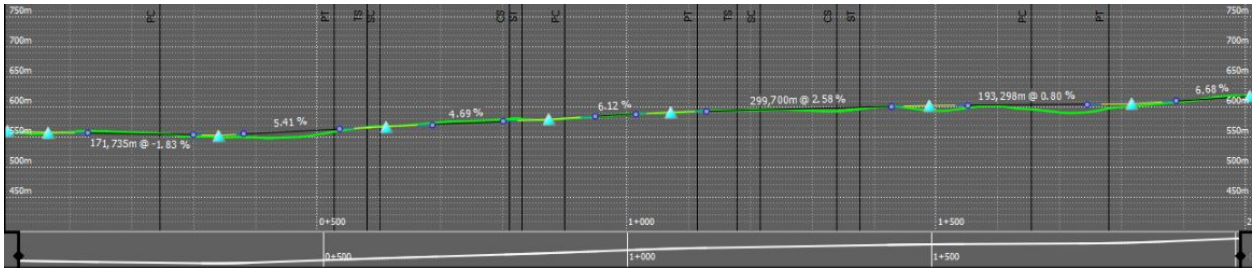


Figure 8 - Vertical alignment of Variant 2

Variant 1 vertical profile implies a less consuming earthmoving, and indeed more viable, as its elevation doesn't converge much with the one of the terrain itself. From both variants, we

consider Variant 2 to be the most viable to construct, most because it doesn't provoke the so called *wounds* in the terrain.

The largest gradient is 6.68% and the minimum is 0.80%.

Cut	35298.804 m^3
Fill	150090.660 m^3
Net Fill	114791.857 m^3

Table 17 - Variant 2 earthwork quantities

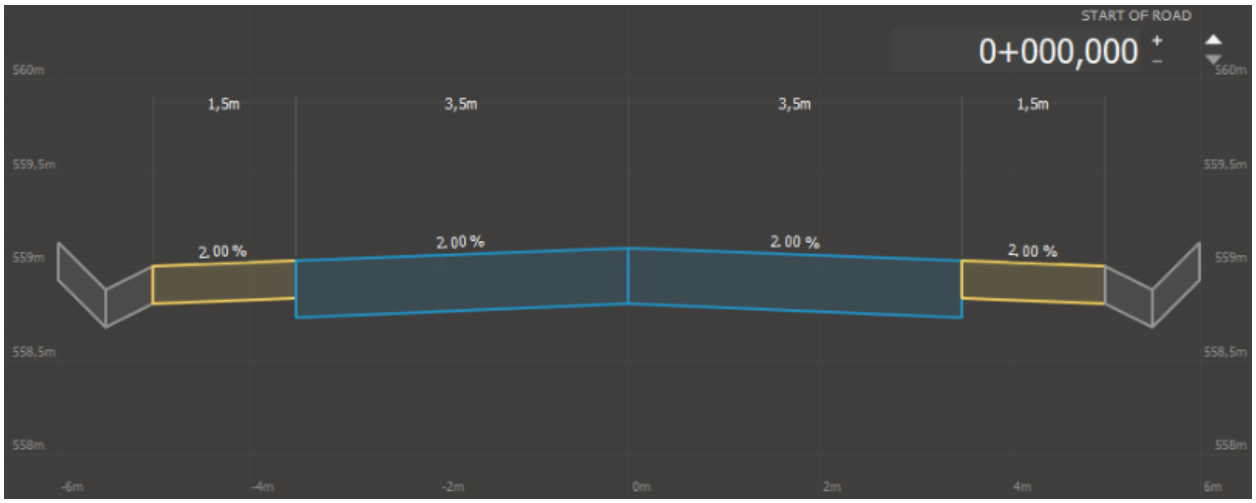
The option for Profile optimization was used, and can be seen in the files in annex (Variants 1.2 and 2.2). While it does provide a much economical earthworks, it has been proving difficult to respect the criteria imposed by the length of high gradients.

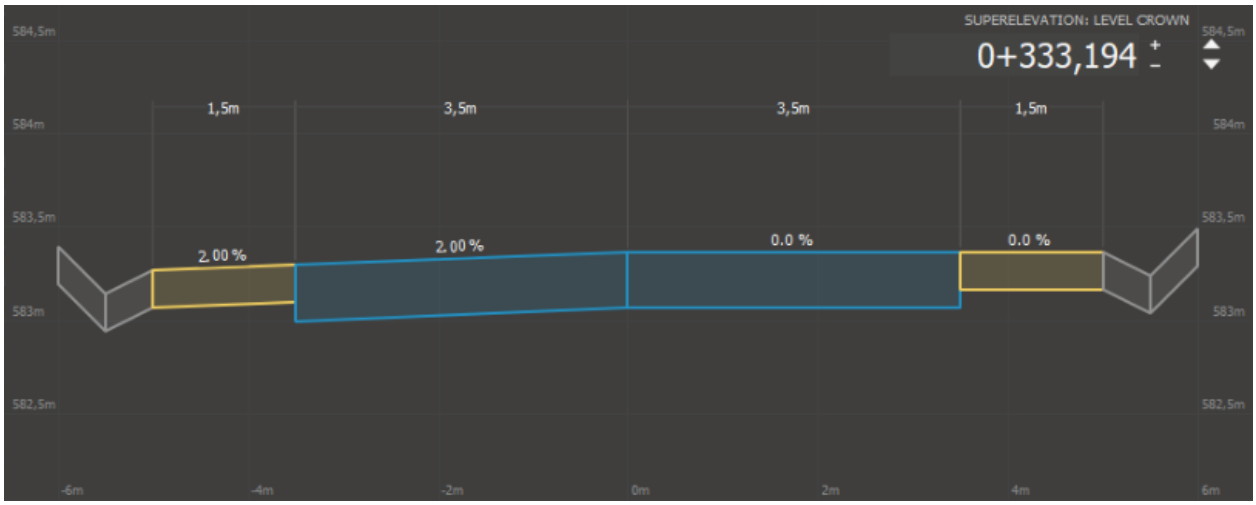
Cross-sectional profile

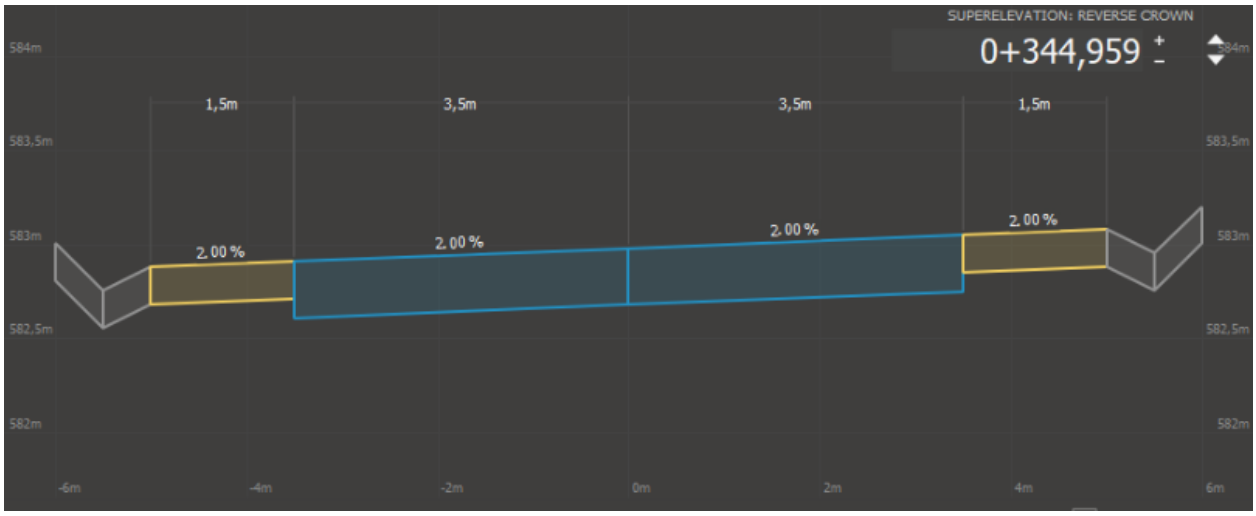
Both profiles used the same values, as mentioned previously. The generalized option for superelevation in Autodesk Infraworks was used, using for:

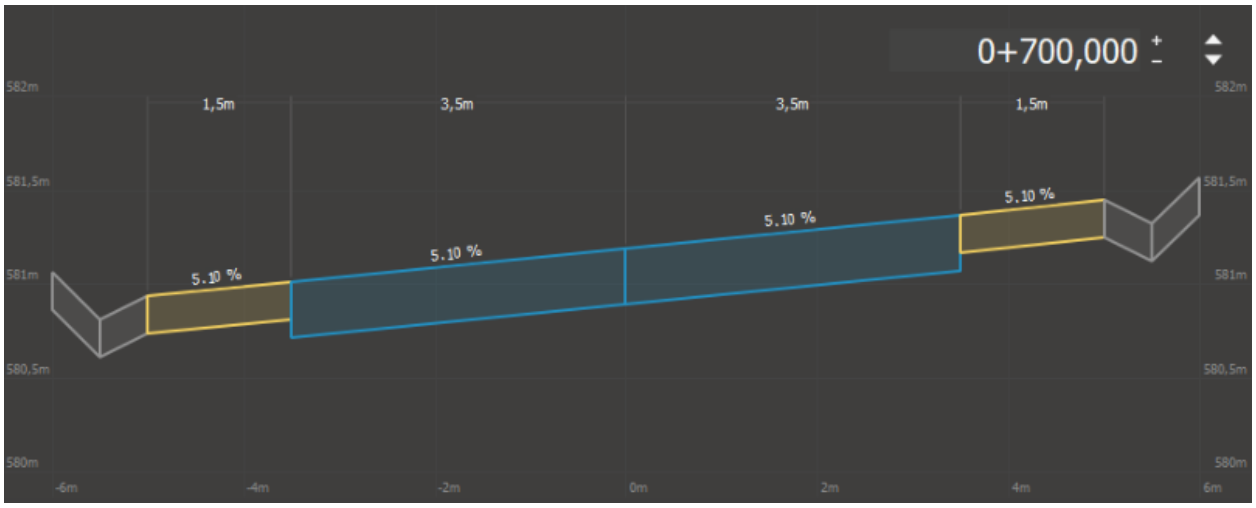
Maximum superelevation rate	7%
Runoff on Tangent	70%
Runoff on Spiral	100%

Example - Variant 1









Discussion

For the reasons we have mentioned previously, the most suitable trace for this road seems to be Variant 2. It fits the criteria and the purpose it was given, though there are still many things to apply and take into consideration (such as barriers and intersections).

One of the criteria we deemed the most important was to preserve the actual terrain, and refrain from making significant cuts in the mountains. The landscape is one of the main reasons that brings visitors to this part of Portugal, and we tried to respect it as much as we could.

Conclusion

Through the course of this project, we were confronted with the hardships of planning and designing a road, and keeping in consideration its application. There were many other things we would like to see applied, but we opted not to take them into consideration.

We believe that we will be better prepared and that we understood better the reality of planning road, contrasting a lot with the theoretical approach we had in our home university.

References

1. Instituto da Mobilidade e dos Transportes (2010). Norma de Traçado;
2. Instituto da Mobilidade e dos Transportes (2010). Barreiras New-Jersey com valeta adjacente: condições e parâmetros de segurança.